An AI Ethical Framework for Data-Centric Engineering

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Abstract

The rapid integration of artificial intelligence into safety-critical engineering systems reveals a key regulatory gap: current codes of conduct and ethical principles for AI fall short when applied to the case where failure of the algorithm leads to real-world risks such as physical, clinical, or infrastructural. This article proposes an ethical framework, geared towards engineers, that seeks to solve this problem in the particular setting of Data-Centric Engineering (DCE). The framework consists of five engineering principles, Verification, Traceability, Ultimate Liability, Boundary Clarity, and Stewardship, all of which are professional imperatives binding the practising engineer rather than the algorithm. In order to ensure that these principles do not remain in the realm of philosophy, they are put into practice through three phases of the engineering process: an algorithm risk gate at the onset of the project, compliance checks integrated within CI/CD pipelines, and a decoupled run-time architecture that ensures deterministically safe overrides. A pre-deployment audit checklist is provided for this purpose. The overarching idea here is that professional accountability cannot be delegated to anything other than a human.

1 Introduction and Context

1.1 Background and Institutional Context

This framework represents one of the key outputs of this research project, providing a formal ethical framework for the use of AI in Data-Centric Engineering (DCE). It serves as an authoritative model providing a baseline for ensuring that deployed computational pipelines are verifiable, transparent, and traceable.. Taking into account the outstanding speed of current technological advancements, it is crucial to acknowledge that the proposed framework is designed to function as a living document, which can be iterated upon dynamically. It is critical to take preemptive measures due to the emergence of socio-technical risks associated with new types of engineering.

1.2 The Engineering Gap in AI Ethics

While established frameworks such as the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems [1] and ISO/IEC 42001 [2] provide essential governance structures for data fairness and AI management systems, they are rooted in a software-centric paradigm. They do not address the cyber-physical realities codified in traditional engineering safety standards such as IEC 61508 [3] or ISO 13849 [4]. Furthermore, professional engineering codes of ethics—such as those of the ASME [5] and ASCE [6]—ground liability in licensed professional practice, a principle this framework extends to AI pipeline authorship. It is the aim of the proposed framework to bridge this vital regulatory gap.

1.3 Terminology and Definitions

The following terms are used with specific technical meaning throughout this document. Precise definitions are necessary to prevent the semantic ambiguity that commonly undermines the practical application of ethical frameworks in engineering contexts.

Artificial Intelligence (AI). As used in this framework, AI denotes any computational system that employs a statistical model, trained on empirical data, to perform prediction, classification, optimisation, or decision-support tasks. This definition explicitly excludes deterministic, rule-based algorithms, which carry conventional engineering liability without the additional epistemic risks addressed here.

Data-Centric Engineering (DCE). An engineering paradigm in which the data pipeline, its provenance, representativeness, and integrity, is treated as the primary engineering artefact, rather than the model architecture itself. DCE recognises that the performance and safety envelope of any AI system is fundamentally bounded by the quality and scope of its training data.

Algorithmic Pipeline. The end-to-end sequence of computational processes through which raw input data is transformed into a model output or engineering recommendation. This encompasses data ingestion, pre-processing, model inference, post-processing, and any downstream actuation or advisory stages.

Out-of-Distribution (OOD) Data. Input data whose statistical characteristics fall outside the distribution represented by the model's training dataset. OOD inputs invalidate the implicit assumptions of a trained model and constitute a primary and frequently underestimated failure mode in deployed AI systems [7].

Human-in-the-Loop (HITL). A system architecture or operational protocol in which a qualified engineer is required to review, validate, or explicitly approve algorithmic outputs before consequential action is taken. HITL mechanisms are the primary instrument through which nondelegable professional liability is preserved at runtime.

Continuous Integration / Continuous Deployment (CI/CD). A software engineering practice in which code changes are automatically compiled, tested, and validated against a defined suite of checks prior to integration or deployment. Within this framework, the CI/CD pipeline serves as the enforcement mechanism for automated ethical and physical boundary compliance checks.

Failure Mode and Effects Analysis (FMEA). A structured, systematic methodology for enumerating the potential failure modes of a system, assessing their root causes and downstream consequences, and quantifying associated risk [8]. This framework extends the conventional FMEA to encompass probabilistic and adversarial failure modes specific to AI systems, referred to herein as a *model-focused FMEA*.

2 The Five Core Engineering Principles

Principle I - Verification

You shall not accept any algorithmic predictions or outputs as being verified truths. All data models, sensitivities, optimisations, and simulations must be constrained within known engineering, statistical, and physical limits, and rigorously validated against empirical reality before use.

Principle II - Traceability

Provenance of training datasets and generated data must be evident to ensure the pipeline is fully reproducible and auditable. For third-party or open-source models where raw provenance is inaccessible, a formal secondary data attestation documenting known dataset limitations is required. Misrepresenting the source of data, concealing algorithmic bias, or promoting unverified AI-generated content as your own engineering design is unethical.

Principle III - Ultimate Liability

The algorithm does not possess professional liability; only the engineer does. You have ultimate, nondelegable liability for the physical and clinical failure modes of any system powered by an AI pipeline if you certify it.

Principle IV - Boundary Clarity

It is your responsibility to define explicitly where the training distribution ends and extrapolation begins using mathematically defensible metrics (e.g., conformal prediction intervals or reconstruction error thresholds). You must foster an environment in which model anomalies, distributional drift, and OOD conditions are systematically detected and disclosed.

Principle V - Stewardship

In fields where technology is advancing rapidly, you must anticipate future failure modes, adversarial vulnerabilities, and the potential for misuse or dual-use of your systems. Where there exists a credible risk of serious harm to human beings or the natural environment, the precautionary principle shall govern.

3 Operationalizing the Framework

To avoid the danger of keeping this framework purely philosophical in nature, it is imperative that these principles be operationalized into concrete engineering processes. This involves incorporating verification checkpoints into the technical system.

3.1 Proportional Governance Using Tiers of Risk

To ensure institutional feasibility, compliance constraints are applied proportionally via a three-tiered risk framework rather than a blanket mandate.

Table 1: Data-Centric Engineering Risk Classification and Governance Matrix

Risk Tier	Operational Boundary & Impact	Mandatory Compliance Requirements
Tier 1: Low	Exploratory research, basic script development, or isolated statistical analysis with no downstream cyber-physical or clinical deployment.	Exempt from formal gating. Adherence to standard version control and documented data sources is encouraged.
Tier 2: Medium	Computational optimization, generative design tools, or advisory simulators where code outputs heavily inform engineering designs but do not actuate systems directly.	Secondary Data Attestation; empirical baseline validation; documented statistical margins of error; internal peer review.
Tier 3: High	Safety-critical automated systems, real-world deployments, clinical decision software, or workflows with direct physical actuation risks.	Full Algorithmic Risk Gate; automated CI/CD adversarial testing; decoupled safety-loop execution; Principal Investigator (PI) sign-off.

3.2 Integration into the Engineering Lifecycle

This framework instantiates a multi-layered governance model that fundamentally departs from conventional, retrospective auditing paradigms. Rather than serving as a post-hoc diagnostic tool, ethical compliance is operationalized as an active, continuous gating mechanism embedded throughout the entire engineering lifecycle across three sequential phases.

Project Inception. Operational controls occur even before the architecture design. Each data-centric engineering pipeline must clear an established project inception gate where a strict algorithmic risk analysis will be conducted. The reason for doing so at the pre-development stage is that this stage clearly defines the operational scope of the system and maps out failure modes and areas to mitigate risk.

Continuous Automated CI/CD Integration. In the process of development, however, these ethical requirements translate to automated checking mechanisms that are intrinsically part of the CI/CD process. In addition to the regular deterministic unit tests, automated test processes conduct continuous adversarial testing, data lineage verification, and mathematical boundary checks. If a new version of a model breaches the physical boundary laws set forth or shows statistical drift from its base distribution, then it will not compile and will not be deployed.

Decoupled System Architecture. The deployment phase mandates a structurally decoupled architecture for runtime execution of safety-critical systems. In this architecture, the predictive AI pipeline runs purely as an advisory system, being subservient to the deterministic, non-probabilistic physical safety loop always. When out-of-distribution (OOD) data inputs or anomalies in optimization process occur, the system initiates the immediate cut-off function. This is because of the structural design feature that causes an instant fallback of the physical asset into a predetermined deterministically safe mode along with the activation of HITL override requiring engineering approval before resumption of autonomy.

3.3 Academic Culture and Research Training

Despite these measures, technical countermeasures will always be insufficient without a complete cultural change in institutional infrastructure that will encourage the development of algorithmic skepticism. The entire university coursework and training protocols should condition researchers

to see machine-learning outputs not as immutable facts, but as extremely complex conditional probability distributions that need to be validated.

4 Conclusion

Accountability cannot be outsourced. Automated coding and AI-assisted tooling can accelerate development, but no tool can substitute the professional responsibility of ensuring a system is safe, validated, and fit for deployment in safety-critical environments. It is the duty of an engineer to enforce strict boundary constraints, explicit data provenance, clear boundary controls, and unyielding human professional liability. By doing so, this encourages the safe and effective usage of AI without compromising the fundamental dedication to public safety and physical engineering excellence.

References

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